

SQL WHERE Clause and Operators - Full Notes

This PDF explains SQL WHERE clause concepts with complete code examples, beginner-friendly explanations, outputs, and operator understanding.

Full SQL Code

```
-- Create Database
CREATE DATABASE college_db;

-- Use Database
USE college_db;

-- Create Table
CREATE TABLE students (
    id INT,
    name VARCHAR(50),
    city VARCHAR(50),
    marks INT
);

-- Insert Data
INSERT INTO students VALUES
(1, 'Ram', 'Hyderabad', 90),
(2, 'Sam', 'Delhi', 75),
(3, 'Ravi', 'Chennai', 60),
(4, 'Rahul', 'Hyderabad', 85);

-- Display Table
SELECT * FROM students;

-- WHERE with Greater Than
SELECT * FROM students
WHERE marks > 70;

-- WHERE with Equal
SELECT * FROM students
WHERE city = 'Delhi';

-- WHERE with AND
SELECT * FROM students
WHERE marks > 80 AND city = 'Hyderabad';

-- WHERE with OR
SELECT * FROM students
WHERE city = 'Delhi' OR city = 'Chennai';

-- WHERE with LIKE
SELECT * FROM students
WHERE name LIKE 'R%';

-- WHERE with BETWEEN
SELECT * FROM students
WHERE marks BETWEEN 70 AND 90;

-- WHERE with IN
SELECT * FROM students
WHERE city IN ('Delhi', 'Hyderabad');
```

1. CREATE DATABASE

CREATE DATABASE creates a new database named college_db.

2. USE DATABASE

USE selects the database to work with.

3. CREATE TABLE

CREATE TABLE creates a table named students with columns id, name, city, and marks.

4. INSERT INTO

INSERT INTO adds rows of data into the students table.

5. SELECT *

SELECT * displays all rows and columns from the table.

6. WHERE marks > 70

Filters students whose marks are greater than 70.

7. WHERE city = 'Delhi'

Displays students whose city is Delhi.

8. AND Operator

AND requires both conditions to be true.

9. OR Operator

OR requires at least one condition to be true.

10. LIKE Operator

LIKE is used for pattern matching. 'R%' means names starting with R.

11. BETWEEN Operator

BETWEEN selects values inside a range.

12. IN Operator

IN checks multiple values in a single condition.

Important SQL Operators

Operator	Meaning
=	Equal
>	Greater Than
<	Less Than
AND	Both conditions must be true

OR	Any one condition can be true
LIKE	Pattern matching
BETWEEN	Range checking
IN	Checks multiple values

Final Summary

The WHERE clause is one of the most important SQL concepts. It helps filter rows using conditions and operators. These concepts are used in:

- Backend Development
- Data Analysis
- Full Stack Development
- AI/ML Projects
- Placements and Interviews